

Class Test: Structure of the Atom (Case Study Based)**Class:** IX**Subject:** Science**Time:** 60 Minutes**Chapter:** Structure of Atom**Maximum Marks:** 40**General Instructions**

- All questions are **compulsory**.
- The question paper consists of **20 case-study based questions**.
- Each question carries **1–3 marks** depending on difficulty.
- Diagrams must be neat and properly labelled wherever required.
- Use of calculator is **not permitted**.
- Atomic masses should be taken from the periodic table where required.

Case Study 1 (1 Mark)

An element contains particles having negligible mass but negative charge.

Question:

Identify the particle described above.

Ans.: _____

Case Study 2 (1 Mark)

A student claims that atoms are indivisible because they cannot be seen.

Question:

Which scientist's model supports this claim?

Ans.: _____



Case Study 3 (2 Marks)

During an experiment, gold foil was bombarded with α -particles. Most passed through undeflected.

Questions:

- a) Name the scientist who performed this experiment. (1)
- b) What conclusion was drawn regarding atomic structure? (1)

Ans(a).: _____

Ans(b).: _____

Case Study 4 (2 Marks)

An atom has 11 protons and 12 neutrons.

Questions:

- a) Write the atomic number. (1) _____
- b) Write the mass number. (1) _____

Ans(a).: _____

Ans(b).: _____

Case Study 5 (2 Marks)

A neutral atom X loses one electron during a chemical reaction.



**Questions:**

- a) What type of ion is formed? (1)
- b) What happens to the charge on the atom? (1)

Ans(a).: _____

Ans(b).: _____

Case Study 6 (2 Marks)

Two atoms A and B have the same mass number but different atomic numbers.

Questions:

- a) What are such atoms called? (1)
- b) Give one example. (1)

Ans(a).: _____

Ans(b).: _____

Case Study 7 (2 Marks)

Atoms of hydrogen, deuterium, and tritium behave similarly in chemical reactions.

Questions:

- a) What is the collective name for these atoms? (1)
- b) Why do they show similar chemical properties? (1)

Ans(a).: _____



Ans(b).: _____

Case Study 8 (2 Marks)

An atom has electronic configuration 2, 8, 1.

Questions:

- a) Identify the element. (1)
- b) Predict whether it is reactive or stable. Give reason. (1)

Ans(a).: _____

Ans(b).: _____

Case Study 9 (2 Marks)

A student observes that the nucleus occupies very little space compared to the atom.

Questions:

- a) Which experiment led to this observation? (1)
- b) What does it indicate about atomic structure? (1)

Ans(a).: _____

Ans(b).: _____



Case Study 10 (2 Marks)

An element has atomic number 17.

Questions:

- a) Write the number of electrons present. (1)
- b) Write its valency with justification. (1)

Ans(a).: _____

Ans(b).: _____

Case Study 11 (3 Marks – Tough)

An ion has 18 electrons and 17 protons.

Questions:

- a) Identify the ion. (1)
- b) Write its electronic configuration. (1)
- c) State whether it is an anion or cation. (1)

Ans(a).: _____

Ans(b).: _____

Ans(c).: _____



Case Study 12 (3 Marks – Tough)

Three elements P, Q, and R have atomic numbers 10, 11, and 12 respectively.

Questions:

- Write electronic configuration of each. ($1\frac{1}{2}$)
- Identify the noble gas among them. ($\frac{1}{2}$)
- Which element is most reactive? Give reason. (1)

Ans(a).: _____

Ans(b).: _____

Ans(c).: _____

Case Study 13 (3 Marks – Tough)

An element has mass number 24 and atomic number 12.

Questions:

- Find the number of neutrons. (1)
- Write its electronic configuration. (1)
- Predict its valency. (1)

Ans(a).: _____

Ans(b).: _____

Ans(c).: _____



Case Study 14 (3 Marks – Tough)

Two ions A^{2+} and B^{-} have the same number of electrons.

Questions:

- Which ion has more protons? (1)
- Give reason for your answer. (1)
- Name the type of bonding likely formed between A and B. (1)

Ans(a).: _____

Ans(b).: _____

Ans(c).: _____

Case Study 15 (3 Marks – Higher Order)

An atom X has 2 electrons in its outermost shell and atomic number 20.

Questions:

- Write its electronic configuration. (1)
- Predict the ion it forms. (1)
- Write the formula of its oxide. (1)

Ans(a).: _____

Ans(b).: _____

Ans(c).: _____



Case Study 16 (3 Marks – HOTS)

An atom forms X^{2-} ion and has 18 electrons in the ion.

Questions:

- a) Find atomic number of the element. (1)
- b) Identify the element. (1)
- c) State its position (group) in the periodic table. (1)

Ans(a).: _____

Ans(b).: _____

Ans(c).: _____

Case Study 17 (3 Marks – HOTS)

An element has configuration 2, 8, 7.

Questions:

- a) How many electrons will it gain or lose? (1)
- b) Write the formula of ion formed. (1)
- c) Justify the formation of this ion. (1)

Ans(a).: _____

Ans(b).: _____

Ans(c).: _____



**Case Study 18 (3 Marks – Conceptual)**

Thomson's atomic model failed to explain certain observations.

Questions:

- a) State one limitation of Thomson's model. (1)
- b) Which experiment disproved it? (1)
- c) Name the improved model proposed later. (1)

Ans(a).: _____

Ans(b).: _____

Ans(c).: _____

Case Study 19 (3 Marks – Numerical Logic)

An atom has equal number of electrons and neutrons, and its mass number is 40.

Questions:

- a) Find atomic number. (1)
- b) Write electronic configuration. (1)
- c) Identify the element. (1)

Ans(a).: _____

Ans(b).: _____

Ans(c).: _____



Case Study 20 (3 Marks – Competitive Level)

An element exists in three forms with mass numbers 35, 37, and 36 but identical chemical properties.

Questions:

- What are these forms called? (1)
- Why do they show same chemical properties? (1)
- Name the element. (1)

Ans(a).: _____

Ans(b).: _____

Ans(c).: _____

End of Question Paper

Name and WhatsApp Number: _____



Scene for Solution

